

Drone Arm A Finite Element Analysis Results

Drone Arm Material	Mass of Drone Arm (kg)	Yield Strength (Pa)	Von Mises Strength (Pa)	Maximum Displacement (μm)	Factor of Safety
Carbon Fiber Composite (CFRP)	0.0207	600000000	1.07E+07	158.773	55.917
Aluminum Alloy	0.0348	275000000	1.07E+07	160.9833	25.6428
Fiberglass Composite (GFRP)	0.0245	300000000	1.07E+07	444.7197	27.9488
PLA Plastic	0.0161	600000000	1.07E+07	3171.7	5.5982
ABS Plastic	0.0136	400000000	1.07E+07	5551.7	3.7313
Wood (Birch)	0.0077	800000000	1.07E+07	1110.3	7.4627

Drone Arm B Finite Element Analysis Results

Drone Arm Material	Mass of Drone Arm (kg)	Yield Strength (Pa)	Von Mises Strength (Pa)	Maximum Displacement (μm)	Factor of Safety
Carbon Fiber Composite (CFRP)	0.0225	600000000	6.91E+05	8.1432	867.7849
Aluminum Alloy	0.0379	275000000	6.91E+05	8.2516	397.767
Fiberglass Composite (GFRP)	0.0267	300000000	6.91E+05	22.8164	433.9205
PLA Plastic	0.0175	600000000	6.91E+05	162.4599	86.814
ABS Plastic	0.0147	400000000	6.91E+05	284.4359	57.8679
Wood (Birch)	0.0084	800000000	6.91E+05	56.8872	115.7358

Drone Arm C Finite Element Analysis Results

Drone Arm Material	Mass of Drone Arm (kg)	Yield Strength (Pa)	Von Mises Strength (Pa)	Maximum Displacement (μm)	Factor of Safety
Carbon Fiber Composite (CFRP)	0.046	600000000	2.76E+05	1.7271	2.17E+03
Aluminum Alloy	0.0777	275000000	2.75E+05	1.7457	1.00E+03
Fiberglass Composite (GFRP)	0.0547	300000000	2.77E+05	4.8466	1.08E+03
PLA Plastic	0.036	600000000	2.73E+05	34.2669	219.443
ABS Plastic	0.0302	400000000	2.73E+05	60.0575	146.3753
Wood (Birch)	0.0173	800000000	2.73E+05	12.0115	292.7505